

Calculus I, Fall 2009

Number	Due	Problems
1	9/16	1.1.18, 1.3.56, 2.2.22, 2.2.38 (a) (b) (c)
2	9/23	2.3.36, 3.3.40, [1.5.32], 2.5.48
3	9/30	2.6.27, [1.6.34], 2.7.8, 2.7.42
4	10/7	2.8.30 (a)(b)(c), 3.1.60, 3.2.50
5	10/21	3.4.66, 3.5.42, 3.9.32, 3.9.38
6	10/28	3.10.34, 3.10.44, 1.6.73 (a), 3.4.82 (a) (b)
7	11/4	3.8.10, 3.8.16, [3.5.32], [3.5.40]
8	11/11	4.4.80, 4.1.14, [4.1.44], 4.1.78
9	11/18	[4.2.16], 4.2.36, 4.3.66, 4.3.68
10	11/25	Two problems from WebAssign (see note), 4.9.50, 4.9.76
11	12/2	4.7.2, 4.7.12, 4.7.46, 4.7.58
12	12/9	5.1.16, [5.2.10], 5.2.58, 5.4.48
13	12/14	5.3.14, 5.3.76, 5.5.34, 5.5.74

[1.5.32]

32. Prove that $\lim_{x \rightarrow 0^+} \sqrt{x} [1 + \sin^2(2\pi/x)] = 0$.

[1.6.34]

34. $F(x) = \frac{x - 9}{\sqrt{4x^2 + 3x + 2}}$

[3.5.32]

32. Find y' if $\tan^{-1}(xy) = 1 + x^2y$.

[3.5.40]

40. A lighthouse is located on a small island, 3 km away from the nearest point P on a straight shoreline, and its light makes four revolutions per minute. How fast is the beam of light moving along the shoreline when it is 1 km from P ?

[4.1.44],

44. $f(x) = \frac{x}{x^2 + 4}$, $[0, 3]$

[4.2.16],

16. Let $f(x) = (x + 1)/(x - 1)$. Show that there is no value of c such that $f(2) - f(0) = f'(c)(2 - 0)$. Why does this not contradict the Mean Value Theorem?

[5.2.10],

10. The table gives the values of a function obtained from an experiment. Use them to estimate $\int_0^6 f(x) dx$ using three equal subintervals with (a) right endpoints, (b) left endpoints, and (c) midpoints. If the function is known to be a decreasing function, can you say whether your estimates are less than or greater than the exact value of the integral?

x	0	1	2	3	4	5	6
$f(x)$	9.3	9.0	8.3	6.5	2.3	-7.6	-10.5

Calculus III, Fall 2009

Number	Due	Problems
1	9/16	12.1.28, 12.1.32, 12.1.40
2	9/23	12.2.40, 12.3.44, 12.4.44, 12.4.46
3	9/30	12.5.64, 12.5.76, [10.6.16], 12.6.50
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[10.6.16],

11–20 ■ Find the traces of the given surface in the planes $x = k$, $y = k$, $z = k$. Then identify the surface and sketch it.

11. $4x^2 + 9y^2 + 36z^2 = 36$

12. $4y = x^2 + z^2$

13. $y^2 = x^2 + z^2$

14. $z = x^2 - y^2$

15. $-x^2 + 4y^2 - z^2 = 4$

16. $25y^2 + z^2 = 100 + 4x^2$

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